Zadanie 4 . Inverzná kinematická úloha pre 6DOF robotický manipulátor

Optimalizácia a optimálne riadenie

- Uvažujme šesťčlánkový 6DOF robotický manipulátor
- Uvažujme polohový vektor $p = \begin{pmatrix} p^x \\ p^y \\ p^z \end{pmatrix}$ a vektor kĺbových premenných $\theta = \begin{pmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \\ \theta_4 \\ \theta_5 \\ \theta_6 \end{pmatrix}$
- Uvažujme nasledovné rotačné matice a translačné vektory

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix} R_y(\theta) = \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{pmatrix} R_z(\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$T_1 = \begin{pmatrix} 0 \\ 0 \\ L_1 \end{pmatrix}, T_2 = \begin{pmatrix} 0 \\ 0 \\ L_2 \end{pmatrix}, T_3 = \begin{pmatrix} 0 \\ 0 \\ L_3 \end{pmatrix}, T_4 = \begin{pmatrix} 0 \\ 0 \\ L_4 \end{pmatrix}, T_5 = \begin{pmatrix} 0 \\ 0 \\ L_5 \end{pmatrix}, T_6 = \begin{pmatrix} 0 \\ 0 \\ L_6 \end{pmatrix}$$

- Kinematika koncového efektora je potom daná ako (transformácia z p_6 do p_0)

$$p_0(\theta) = R_z(\theta_1)\left(T_1 + R_x(\theta_2)\left(T_2 + R_x(\theta_3)\left(T_3 + R_z(\theta_4)\left(T_4 + R_y(\theta_5)\left(T_5 + R_z(\theta_6)(T_6)\right)\right)\right)\right)\right)$$

- Jakobián polohového vektora $p_0(\theta)$ možno definovať ako maticu parciálnych derivácii $\nabla p_0(\theta) \in \mathbb{R}^{3\times 6}$

$$\nabla p_0(\theta) = \left(\frac{\partial p_0(\theta)}{\partial \theta_1} \quad \frac{\partial p_0(\theta)}{\partial \theta_2} \quad \frac{\partial p_0(\theta)}{\partial \theta_3} \quad \frac{\partial p_0(\theta)}{\partial \theta_4} \quad \frac{\partial p_0(\theta)}{\partial \theta_5} \quad \frac{\partial p_0(\theta)}{\partial \theta_6}\right)$$

- Nájdime teraz parciálne derivácie podľa kĺbových premenných

$$\frac{\partial p_0(\theta)}{\partial \theta_1} = \frac{\partial\left(R_z(\theta_1)\right)}{\partial \theta_1}\left(T_1 + R_x(\theta_2)\left(T_2 + R_x(\theta_3)\left(T_3 + R_z(\theta_4)\left(T_4 + R_y(\theta_5)\left(T_5 + R_z(\theta_6)(T_6)\right)\right)\right)\right)\right)$$

$$\frac{\partial p_0(\theta)}{\partial \theta_2} = R_z(\theta_1)\left(\frac{\partial\left(R_x(\theta_2)\right)}{\partial \theta_2}\left(T_2 + R_x(\theta_3)\left(T_3 + R_z(\theta_4)\left(T_4 + R_y(\theta_5)\left(T_5 + R_z(\theta_6)(T_6)\right)\right)\right)\right)\right)$$

$$\frac{\partial p_0(\theta)}{\partial \theta_3} = R_z(\theta_1)R_x(\theta_2)\left(\frac{\partial\left(R_x(\theta_3)\right)}{\partial \theta_3}\left(T_3 + R_z(\theta_4)\left(T_4 + R_y(\theta_5)\left(T_5 + R_z(\theta_6)(T_6)\right)\right)\right)\right)$$

$$\frac{\partial p_0(\theta)}{\partial \theta_4} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)\left(\frac{\partial\left(R_z(\theta_4)\right)}{\partial \theta_4}\left(T_4 + R_y(\theta_5)\left(T_5 + R_z(\theta_6)(T_6)\right)\right)\right)$$

$$\frac{\partial p_0(\theta)}{\partial \theta_5} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)\left(\frac{\partial\left(R_y(\theta_5)\right)}{\partial \theta_5}\left(T_5 + R_z(\theta_6)(T_6)\right)\right)$$

$$\frac{\partial p_0(\theta)}{\partial \theta_6} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)R_y(\theta_5)\frac{\partial\left(R_z(\theta_6)\right)}{\partial \theta_6}(T_6)$$

- Rotačné matice jednotlivých článkov robota tvoria celkovú rotačnú maticu pre súradnicový systém efektora

$$R_T(\theta) = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)R_y(\theta_5)R_z(\theta_6)$$

- Nájdime teraz derivácie $R_T(\theta)$ vzhľadom na kĺbové premenné

$$\frac{\partial R_T(\theta)}{\partial \theta_1} = \frac{\partial R_z(\theta_1)}{\partial \theta_1}R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)R_y(\theta_5)R_z(\theta_6)$$

$$\frac{\partial R_T(\theta)}{\partial \theta_2} = R_z(\theta_1)\frac{\partial R_x(\theta_2)}{\partial \theta_2}R_x(\theta_3)R_z(\theta_4)R_y(\theta_5)R_z(\theta_6)$$

$$\frac{\partial R_T(\theta)}{\partial \theta_3} = R_z(\theta_1)R_x(\theta_2)\frac{\partial R_x(\theta_3)}{\partial \theta_3}R_z(\theta_4)R_y(\theta_5)R_z(\theta_6)$$

$$\frac{\partial R_T(\theta)}{\partial \theta_4} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)\frac{\partial R_z(\theta_4)}{\partial \theta_4}R_y(\theta_5)R_z(\theta_6)$$

$$\frac{\partial R_T(\theta)}{\partial \theta_5} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)\frac{\partial R_y(\theta_5)}{\partial \theta_5}R_z(\theta_6)$$

$$\frac{\partial R_T(\theta)}{\partial \theta_6} = R_z(\theta_1)R_x(\theta_2)R_x(\theta_3)R_z(\theta_4)R_y(\theta_5)\frac{\partial R_z(\theta_6)}{\partial \theta_6}$$

Derivácie rotačných matíc nech sú dané ako:

$$\frac{\partial R_x(\theta)}{\partial \theta} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -\sin(\theta) & -\cos(\theta) \\ 0 & \cos(\theta) & -\sin(\theta) \end{pmatrix} \quad \frac{\partial R_y(\theta)}{\partial \theta} = \begin{pmatrix} -\sin(\theta) & 0 & \cos(\theta) \\ 0 & 0 & 0 \\ -\cos(\theta) & 0 & -\sin(\theta) \end{pmatrix} \quad \frac{\partial R_z(\theta)}{\partial \theta} = \begin{pmatrix} -\sin(\theta) & -\cos(\theta) & 0 \\ \cos(\theta) & -\sin(\theta) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Definujme reziduum (chybu) pre polohu efektora ako

$$r_p(\theta) = p_0(\theta) - p_r$$

- Definujme reziduum (chybu) pre orientáciu efektora ako

$$r_o(\theta) = \begin{pmatrix} (0 & 1 & 0)\left(R_T(\theta){R_R}^T\right)\begin{pmatrix}1\\0\\0\end{pmatrix} \\ (0 & 0 & 1)\left(R_T(\theta){R_R}^T\right)\begin{pmatrix}1\\0\\0\end{pmatrix} \\ (0 & 0 & 1)\left(R_T(\theta){R_R}^T\right)\begin{pmatrix}0\\1\\0\end{pmatrix} \end{pmatrix}$$

- Výsledkom riešenia by malo byť zabezpečiť že všetky reziduá sú nulové

$$r_p(\theta) = \mathbf{0}$$

$$r_o(\theta) = \mathbf{0}$$

- Krok Newtonovej metódy nech spĺňa sústavu lineárnych rovníc

$$\left(\begin{pmatrix} \frac{\partial p_0(\theta)}{\partial \theta_1} & \frac{\partial p_0(\theta)}{\partial \theta_2} & \frac{\partial p_0(\theta)}{\partial \theta_3} & \frac{\partial p_0(\theta)}{\partial \theta_4} & \frac{\partial p_0(\theta)}{\partial \theta_5} & \frac{\partial p_0(\theta)}{\partial \theta_6} \\ (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_1}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_2}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_3}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_4}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_5}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 1\ \ 0)\frac{\partial R_T(\theta)}{\partial \theta_6}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} \\ (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_1}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_2}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_3}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_4}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_5}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_6}{R_R}^T\begin{pmatrix}1\\0\\0\end{pmatrix} \\ (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_1}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_2}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_3}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_4}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_5}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} & (0\ \ 0\ \ 1)\frac{\partial R_T(\theta)}{\partial \theta_6}{R_R}^T\begin{pmatrix}0\\1\\0\end{pmatrix} \end{pmatrix}_{\theta=\theta(k)} + \Lambda\right)\Delta\theta(k) = -\begin{pmatrix} r_p(\theta(k)) \\ r_o(\theta(k)) \end{pmatrix}$$

$$\theta(k+1) = \theta(k) + \Delta\theta(k)$$

- Algoritmus ukončujeme ak je splnená podmienka (veľkosť rezidua)

$$\left(r_p(\theta)\right)^T\left(r_p(\theta)\right) + \left(r_o(\theta)\right)^T\left(r_o(\theta)\right) < \epsilon$$

Implementujte funkciu pre výpočet celkovej rotačnej matice (priečinok robot_arm)

```
function [R_tot] = get_R_tot(theta, R)
R_tot = eye(3,3);

for i = 1:length(R)
    R_tot = %TODO
end

end
```

Implementujte funkciu pre výpočet derivácie celkovej rotačnej matice (priečinok robot_arm)

```
function [dR_tot] = get_dR_tot(theta, R, dR, j)
dR_tot = eye(3);

for i = 1:length(R)
    if i == j
        dR_tot = %TODO
    else
        dR_tot = %TODO
    end
end

end
```

Implementujte funkciu pre výpočet Jakobiánu robota (priečinok robot_arm)

```
function [J] = get_Jacobian(theta, pn, R, dR, T)
J = zeros(3, length(R));

for i = 1:length(R)

   J(:, i) %TODO

end

end
```

Implementujte funkciu pre výpočet priamej kinematickej úlohy

```
function [p0] = transform_pi_to_p0(theta, pi, i, R, T)
p0 = pi;
for j = i:-1:1
```

```
    p0 = %TODO
end

end
```

Implementujte funkciu pre riešenie inverznej kinematickej úlohy (priečinok robot_arm)

```
function [theta] = solve_IK_orientation(pr, Rr, theta, R, dR, T, tol,Lambda)


while true

    p0 = transform_pi_to_p0(theta, [0;0;0], length(R), R, T);
    rp = %TODO

    R_tot = get_R_tot(theta, R);

    Re=%TODO
    ro = %TODO
    Jp = get_Jacobian(theta, [0;0;0], R, dR, T);

    Jo = zeros(3, length(R));

    for i = 1:length(R)
        dR_tot = get_dR_tot(theta, R, dR, i);

        Jo(:, i) = %TODO
    end

    J = %TODO
    delta_theta = %TODO
    theta = %TODO

    if (rp' * rp + ro' * ro) < tol
        break
    end
end


end
```

Použite skript *robot_arm_scr* (priečinok scr)

```
% Define link lengths
T1 = [0; 0; 0.25];
T2 = [0; 0; 0.75];
T3 = [0; 0; 1.0];
T4 = [0; 0; 0.5];
T5 = [0; 0; 0.4];
T6 = [0; 0; 0.3];

% Combine all translation vectors
T = [T1, T2, T3, T4, T5, T6];

R = {@rotZ, @rotX, @rotX, @rotZ, @rotY, @rotZ};
dR = {@drotZ, @drotX, @drotX, @drotZ, @drotY, @drotZ};
```

```
%% Inverse Kinematics with Orientation

% Desired position of end effector
pr = [1.0; 0.5; 2.0];

% Desired orientation using rotation matrices
Rr = rotZ(pi/3) * rotY(pi/3) * rotX(pi/3);

% Initial guess for joint angles

theta =[pi;0;pi/2;pi;pi/2;pi/2];
Lambda=diag([1e-1,1e-1,1e-1,1e-1,1e-1,1e-1]);

% Solve inverse kinematics including orienňtation constraint
theta = solve_IK_orientation(pr, Rr, theta, R, dR, T, 1e-6,Lambda);

% Transform last frame position to base frame
p0 = transform_pi_to_p0(theta, [0;0;0], 6, R, T);

% Plot the resulting robot configuration
plot_robot(theta, R, T)

plot3(pr(1),pr(2),pr(3),'or','LineWidth',2,'MarkerSize',11)

e = Rr * [1;0;0];
quiver3(pr(1), pr(2), pr(3), e(1), e(2), e(3), 0.2, 'r','LineWidth',2);
e = Rr * [0;1;0];
quiver3(pr(1), pr(2), pr(3), e(1), e(2), e(3), 0.2, 'g','LineWidth',2);
e = Rr * [0;0;1];
quiver3(pr(1), pr(2), pr(3), e(1), e(2), e(3), 0.2, 'b','LineWidth',2);
```

Riešenie

